

Exploration of biomarkers for total fish intake in pregnant Norwegian women.

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OBJECTIVE: Few biomarkers for dietary intake of various food groups have been established. The aim of the present study was to explore whether selenium (Se), iodine, mercury (Hg) or arsenic may serve as a biomarker for total fish and seafood intake in addition to the traditionally used n-3 fatty acids EPA and DHA. DESIGN: Intake of fish and seafood estimated by an FFQ was compared with intake assessed by a 4 d weighed food diary and with biomarkers in blood and urine. SETTING: Validation study in the Norwegian Mother and Child Cohort Study (MoBa). SUBJECTS: One hundred and nineteen women. RESULTS: Total fish/seafood intake (median 39 g/d) calculated with the MoBa FFQ was comparable to intake calculated by the food diary (median 30 g/d, $r_s = 0.37$, $P < 0.001$). Erythrocyte DHA and blood Hg, Se and arsenic concentrations were positively correlated with intake of fish and seafood, but the association for DHA was weakened by the widespread use of supplements. The main finding was the consistent positive association between the intake of fish/seafood and blood arsenic concentration. In multivariate analyses, blood arsenic was associated with blood Hg and fish and seafood intake. In these models, arsenic turned out to be the best indicator of intake of fish and seafood, both totally and in subgroups of fish/seafood intake. CONCLUSIONS: While DHA reflected the intake of fatty fish and n-3 PUFA supplements, blood arsenic concentration also reflected the intake of lean fish and seafood. Blood arsenic appears to be a useful biomarker for total fish and seafood intake.